

Michael Adu

Embedded Software Engineer
michaelngadu@gmail.com +447920381515

PROFILES

Github
Michael-Adu

Linkedin
Michael Adu

SUMMARY

A passionate, motivated and project-oriented Embedded Systems Engineer, constantly looking for a challenge and the opportunity to further my knowledge and expand my skills. Skilled in embedded programming languages such as C/C++ and have extensive experience with microcontrollers and SoCs such as Arduinos, Raspberry Pi's, ARM Cortex-M microcontrollers and PIC18 microcontrollers.

EDUCATION

Coventry University	Masters • Distinction
Embedded Systems Engineering	Coventry, United Kingdom • 2022 - 2023
Ashesi University	Bachelors
Electrical and Electronic Engineering	Berekuso, Ghana • 2017-2021

EXPERIENCE

Skyships Automotive Ltd.	Braintree, United Kingdom
Software Engineer	January 2024 - Present
Graduate Embedded Software Engineer	January 2024 - November 2025
• Designed and developed automotive infotainment and cockpit controllers for high end vehicles, integrating CAN messages with various telltales and warnings using QNX.	
• Created several tools to aid in software testing, including a suite of tools to exponentially reduce the duration necessary for calculating and sending automotive CAN messages, performing various critical tests.	
• Developed 2D and 3D user interfaces using Kanzi and Lua.	

Premium Technologies Ltd.	Accra, Ghana
Embedded Engineer and Lead Web Developer	February 2022 - August 2022
• Designed and developed a custom Smart Home System using ESP8022 to interface with already established home electrical systems to be implemented in over 3000 homes.	
• Created a Linux-based Smart Home Hub running on a Raspberry Pi and a companion phone application using Flutter to enable user control of household appliances and monitoring of CCTV cameras on a local network.	

Mingo Blox	Accra, Ghana
Embedded Engineer and Web Developer	June 2021 - February 2022
• Developed a desktop and mobile graphical-based code application for quick and easy programming of Arduino devices for novel engineering students and hobbyists using ReactJS and C/C++, which was marketed to an estimated 30 million users	
• Taught the fundamentals of robotics to 30+ young students at the primary level	

PROJECTS

Automated Sign Language Detection Device	2022-2023
Developing a physical device to recognize sign language hand gestures with the aid of a PIC18F, piezoelectric sensors and machine learning to aid potentially over 430 million audible-impaired individuals with communication.	
Trained a Long Short Term Model Neural Network model to recognize gestures made by the user. Skills used: C/C++, Proteus, Python, PIC18F development, Git and MATLAB.	
Yocto GUI	2022-Present
Developing a graphical user interface for creating Linux Yocto-based operating systems for embedded systems.	
The application aims to bring ease of access to Yocto and provide a centralised workflow when developing operating systems to aid in an industry with an estimated \$116 billion market size.	
Skills used: C/C++, Flutter, NodeJS, Linux	
Range Sensor	2022
Developed a custom Embedded Linux operating system using Yocto to be deployed on a Raspberry Pi with a custom Ultrasonic Sensor device Driver	
Skills used: Yocto, BitBake, C/C++	

SKILLS

Coding
C/C++, Python, VHDL, HTML, Javascript, Dart, PHP, SQL, Lua
Microcontrollers/SoC
Qualcomm, PIC18, STM32, Arduino, Raspberry Pi, ESP32
Communication Protocols
CAN, UART, UDP/TCP, SPI, I2C
Operating Systems
Linux, QNX, RTOS
Code Project Management
Git, Github
Application Frameworks
React, Flutter, QT, PyQt
HMI
Kanzi
Others
Blender, Figma

INTERESTS

3D Modelling

REFERENCES

Available Upon Request